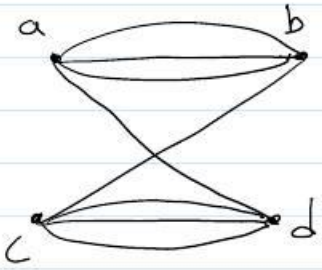
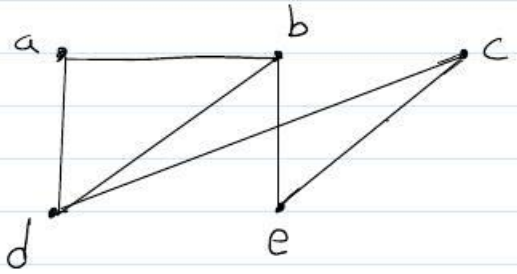
Tutorial 5: Graph Theory II

26 March 2024 23:56

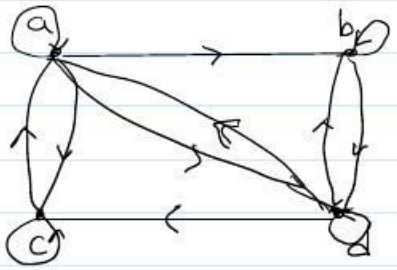
Question 1:



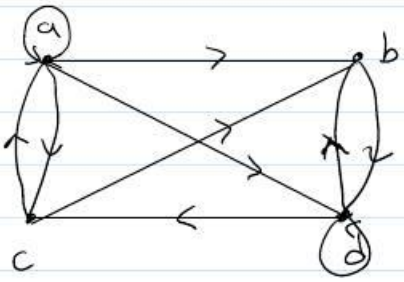
	a	b	c	d
a	0	3	0	1
b	3	0	1	0
c	0	1	0	3
d	1	0	3	0



	a	b	c	d	e
a	0	1	0	1	0
b	1	0	0	1	1
c	0	0	0	1	1
d	1	1	1	0	0
e	0	1	1	0	0



	a	b	c	d
a	1	1	1	1
b	0	1	0	1
c	1	0	1	0
d	1	1	1	1

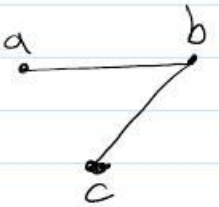


	a	b	c	d
a	1	1	1	1
b	0	0	0	1
c	1	1	0	0
d	0	1	1	1

Question 2:

	a	b	c
a	0	1	0
b	1	0	1
c	0	1	0

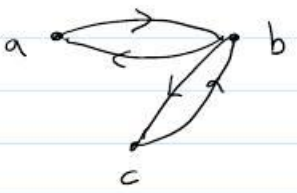
undirected:



directed

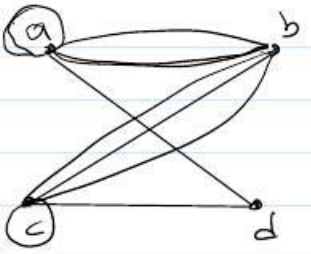


directed



	a	b	c	d
a	1	2	0	1
b	2	0	3	0
c	0	3	1	1
d	1	0	1	0

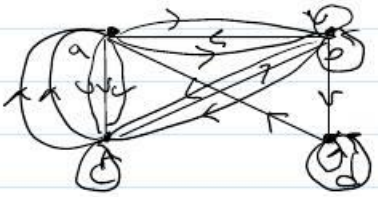
undirected:



directed: duplicate edges and put arrows.

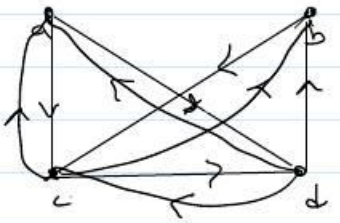
	a	b	c	d
a	0	2	3	0
b	1	2	2	1
c	2	1	1	0
d	1	0	0	2

This matrix is not symmetric $\Rightarrow$ it is a matrix for a directed graph



	a	b	c	d
a	0	0	1	1
b	0	0	1	0
c	1	1	0	1
d	1	1	1	0

Again not symmetric,



Question 3: having zeros on the diagonal makes the graph with no loops. Similarly, a one from $v_i - v_j$ puts only one edge / directed edge on the graph $\Rightarrow$ The Answer is Yes.

Question 4: Notice that $a_{ij} = n$ for $i \neq j$ if v_i is connected to v_j through n edges. Thus, summing these numbers is the same as counting the edges connected to the vertex. Since $a_{ii} = 1$ if $\exists$ a loop on the vertex and a loop contribute 2 to the

counting the edges connected to the vertex since $a_{ii} = 1$ if $\exists$ a loop on the vertex and a loop contribute 2 to the degree of a.

$$\Rightarrow \sum_k a_{ik} = \deg(v) - \text{loops}$$

for undirected graphs

Similarly for directed graphs if the entry is not zero that means that there is a directed edge out of that vertex. So, it contributes to the out degree

$$\Rightarrow \sum_k a_{ik} = \deg^+(v)$$

notice here there is no loops since a loop contributes only to the out degree.

Now for columns, for undirected graphs since $a_{ij} = a_{ji}$ it's symmetric

$$\Rightarrow \sum_i a_{ik} = \sum_i a_{ki} = \deg(v) - \text{loops}$$

Same as above.

However, for directed graphs this contributes to the in degree

$$\Rightarrow \sum_i a_{ik} = \deg^-(v).$$

Question: for K_n every vertex is connected to every other vertex

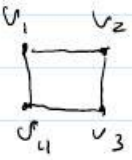
$$\Rightarrow K_n = \begin{bmatrix} 0 & 1 & 1 & 1 & \cdots \\ 1 & \ddots & 1 & 1 & 1 \\ 1 & 1 & \ddots & \ddots & 1 \\ 1 & 1 & 1 & \ddots & 0 \end{bmatrix} \leftarrow \text{so zero on the diagonal and one every where else}$$

$$\Rightarrow a_{ij} = 1 - \delta_{ij} \quad \text{where} \quad \delta_{ij} \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

$\Rightarrow a_{ij} = 1 - \delta_{ij}$ where $\delta_{ij} \begin{cases} 1 & \\ 0 & \text{if } i \neq j \end{cases}$

for C_n . each vertex is connected to the adjacent vertex for example,

C_4



$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Notice that $\forall i$ $a_{i,i+1}$ & $a_{i,i-1}$ are ones this reflects that the vertex is connected to two adjacent vertices.

define $a_{1,-1} = a_{1,n}$ & $a_{n,n+1} = a_{n,1}$

$\Rightarrow$ $a_{ij} = \delta_{i,j+1} + \delta_{i,j-1}$

Question 6: first look for graph invarints if the invarints are different $\Rightarrow$ two graphs are not isomorphic also graph invarint may help us finding the map establishing isomorphism.

invariant	Graph 1	Graph 2
# of vertices	5	5
# of edges	7	7
deg of deg of vertex	$\{u_1, u_2, u_3, u_4, u_5\}$ $\{3, 2, 3, 3, 3\}$	$\{v_1, v_2, v_3, v_4, v_5\}$ $\{2, 4, 2, 3, 3\}$

Notices here the two graphs has two different sequences of degree of vertices sine graph 2 has $\deg(v_2) = 4$ but there is no deg 4 vertex in graph 1 thus, the two graphs are not isomorphic.

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we do the same for the other two graphs,

invarint	graph 1	graph 2
# of vertice	5	5
# of edges	8	8
seq of vertex degree	$\{u_1, u_2, u_3, u_4, u_5\}$ $\{3, 4, 2, 4, 3\}$	$\{v_1, v_2, v_3, v_4, v_5\}$ $\{3, 2, 4, 3, 4\}$

Notice that the two graphs has the same sequance of degress up to some permutation.

Notice that

$u_3 \longrightarrow v_2$ have only degree (2)

however, this sequance has a problem u_3 is connected to two 4 degree vertex while v_2 is connected are 4 degree vertex and on 3 degree. Thus, the two graphs are not isomorphic.

Question 7: Isomorphic graphs must have the same adjancy martix up to a permutation. Thus, we swap corresponding rows/colums to find ismorphism.

a)

$$\begin{array}{c} \\ a \\ b \\ c \end{array}\begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \end{array} \xrightarrow[a, c]{\text{swap}} \begin{array}{c} \\ c \\ b \\ a \end{array}\begin{array}{c} \begin{array}{ccc} c & b & a \end{array} \\ \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{array}$$

Thus, we have reached the other matrix
$\Rightarrow$ yes these graphs are isomorphic.

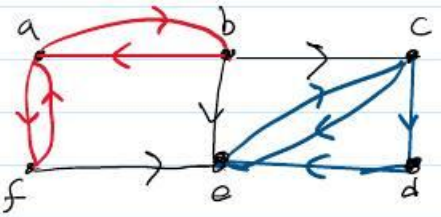
b) Notice these two graphs cannot be ismorphic since in the first matrix only one vertex has degree three, while in the second 2 vertices have degree three.

Question 8: for a graph to be strongly connected $\exists$ a path from $v_i \to v_j$ & $v_j \to v_i$

- $a \to b$ $\{a \to e \to d \to b\}$
- $b \to a$ $\{b \to a\}$
- $a \to c$ $\{a \to e \to d \to c\}$
- $c \to a$ {no out edges from c}

Thus, the graph is not strongly connected however, the graph is weakly connected since the underlying undirected graph is connected.

Question g: To find strongly connected components we search for the largest cycles we can make out of the graph then remove these vertices with edges and search again,



$\{a, b, f\}$, $\{c, e, d\}$ ← These two are strongly connected since they make the biggest loops.

after removing these vertices we would have no other vertices to look at thus, these are the strongly connected components.

Question 10: we will proof a more general statement

- A graph with v vertices and e edges has at least $v - e$ connected components.

proof: by induction on e, If $e = 0$ then we have v isolated points that we have $v - 0 = v$ connected components.

suppose this is true for a graph $G(v, e)$. we want to see how this can affect $G'(v, e+1)$. Choose an arbitrary edge ab in G' and remove it thus, now G' has v vertices and e edges by the induction hypothesis G' has a least $v - (e-1)$ components.

Now, return that edge back. That edge has two cases either ab lies in a connected component or ab connects two different components.

Case 1 : components $\geq v-(e-1)$

Case 2 : components $\geq v-(e-1)-1 = v-e$

in both case components $\geq v-e$ proving the statement.

Now, when components $=1$ (i.e the graph is connected) then we have

$$1 \geq v-e \Rightarrow e \geq v-1$$

Thus, a connected graph has at least $v-1$ edges.